



मौलाना आज़ाद राष्ट्रीय प्रौद्योगिकी संस्थान भोपाल, भोपाल - 462003

Maulana Azad National Institute of Technology Bhopal, Bhopal – 462003

(An Institution of National Importance under Ministry of Education, Govt. of India)

Department of Computer Science & Engineering

B.Tech. V Semester (EE1)

Data Structures (CSE352)

Assignment-I

1. Prove that $f(n) = 7n^2 + 3n + 2$ is $O(n^2)$.
2. Prove that $(n + k)^m = O(n^m)$, where k and m are constants
3. Use Master Theorem for Divide and Conquer Recurrences to prove
 - (a) $T(n) = 16 T(n/4) + n$ is $\Theta(n^2)$
 - (b) $T(n) = 3 T(n/2) + n^2$ is $\Theta(n^2)$.
4. What is the time complexity of the following function:

```
void fun(int n)
{
    int i, j;
    for (i = 1; i <= n ; i++)
    {
        for (j = 1; j < n; j += i)
        {
            printf("%d %d", i, j);
        }
    }
}
```

5. What is the running time of the following function?

```
void Function(int n) {  
    int i=1, s=1;  
    while( s <= n) {  
        i++;  
        s= s+i;  
        printf("*");  
    }  
}
```